Replicator dynamics

The room is the population

Split
Dig or Stay. Stand on your side.

Select
The fitter side gains a few.

Settle
Watch where it stops moving.

In the room: [ ]

Starting split: [ ] digging

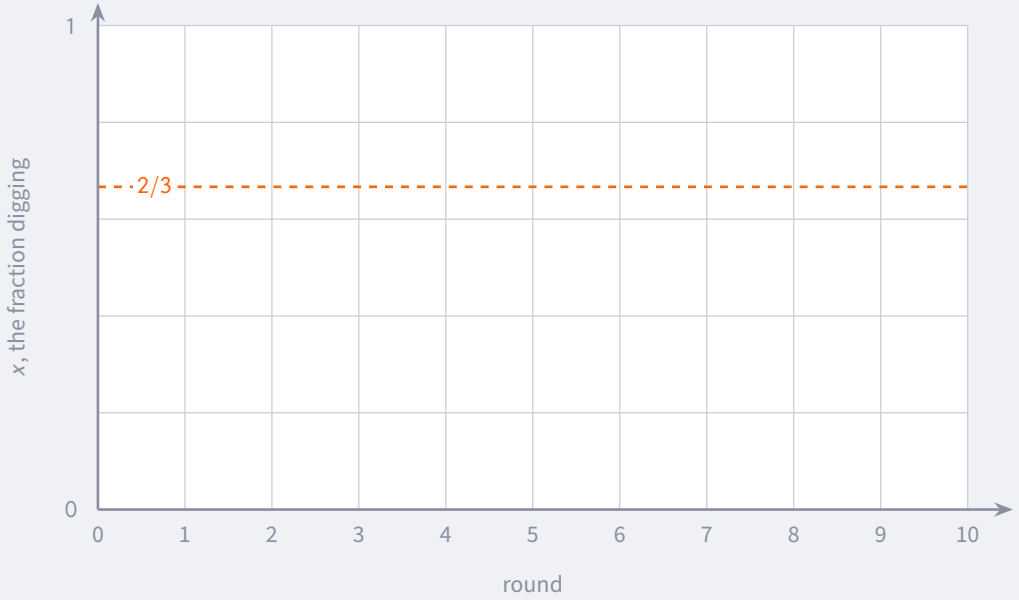
	Dig	Stay
Dig	3	2
Stay	4	0

Clearing the drift is worth 4 to each of you. Digging costs 2, shared if you both dig.

Two diggers get 3 each. A lone digger gets [ ] while the other free-rides for [ ]. Two stayers get [ ].

Would you rather be the one digging? [ ]

Stand on the side of the room for the action you are playing, Dig or Stay. Each round we record the fraction digging and work out which side scored more. Anyone may change sides before the next round.



Round	x	f_D	f_S
1			
2			
3			
4			
5			
6			

Plot the split each round. The dashed line is where we expect it to end up.

With a fraction x digging,

$f_D =$ $\qquad$ $f_S =$

They are equal when $x =$, which is where the room _____.

A strategy grows when it does better than the population average $\phi = xf_D + (1-x)f_S$. How fast it grows depends on two things:

how far above average it is: _____ and how many already play it: _____

$$\dot{x} = _____$$

Each round of the activity, with its small fixed step, was one step of _____ on this equation.

Factorise $\dot{x}$ and read off every value where it is zero:

Rest points: [ ] [ ] [ ]

For $0 < x < 2/3$ the sign of $\dot{x}$ is [ ], and for $2/3 < x < 1$ it is [ ]. So from anywhere inside, the population moves towards [ ].

A small group of Stayers invades a population sitting at $x = 2/3$. Do they earn more or less than the residents?

So $x = 2/3$ _____ an ESS. Rock-Paper-Scissors has an interior rest point too, but there the dynamics _____ instead of settling.

What we just did in the room is **Question 1** on the Replicator Dynamics page.

(a)	The fitnesses f_D and f_S as functions of x.	[3]
(b)	The replicator equation, simplified to a product of factors.	[7]
(c)	All stable populations.	[3]
(d)	From the sign of $\dot{x}$, which one we approach, and where the class settled.	[6]
(e)	Whether the interior population is an ESS, argued two ways.	[6]

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Every stage of today, and the part of the notes it was reading.

What we did	Notes	In the textbook
The snowdrift as a payoff matrix	**10.2.5**	Replicator Dynamics: Definition: Pairwise Interaction Game
How well each side was doing	**10.2.3**	Replicator Dynamics: Definition: Fitness of a strategy in a population
The room as a differential equation	**10.2.1**	Replicator Dynamics: Definition: Replicator Dynamics Equation
Where the fraction settles	**10.2.2**	Replicator Dynamics: Definition: Stable Population
Resisting a small invasion	**10.2.4, 10.2.6**	Replicator Dynamics: Post Entry Population, Evolutionarily Stable Strategy
Examinable	**10.2.7**	Theorem: Characterisation of ESS in two-player games, and proof

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Quick quiz before you go: vknight.org/gt/topics/replicator-dynamics.html

Before the next class: read the Replicator Dynamics chapter as far as the characterisation of ESS, and attempt Question 1.